

IT5008: Tutorial 1 — Entity-Relationship Model

Pratik Karmakar

School of Computing,
National University of Singapore

AY26/27 S1



The Book Exchange Scenario

Students at the National University of Ngendipura (NUN) buy books for their studies, and lend/borrow books to/from each other.

Apasaja Private Limited is commissioned by the **NUN Students Association (NUNStA)** to implement an online book exchange system recording:

- Information about **students**,
- **Books** that they own,
- Books that they **lend and borrow**.

What the Database Must Track

- **Student:** name, main department, identified by email; joined date; graduation date *if* graduated.
- **Department:** belongs to *exactly one* faculty; no faculty exists without a department; some departments have no students.
- **Book:** title, author(s), publisher, language, year, edition, ISBN-10 and ISBN-13. Books may exist unowned (owner graduated, or advised but not yet bought).
- **Copy:** a student may own several copies of the *same* book, told apart by a **copy number** (consecutive from 1).
- **Loan record:** date a copy was borrowed, date it was returned. A student can only borrow/lend *after* being enrolled.
- **Auditing:** keep books/copies/owners as long as owned by a student *or* referenced by a loan; keep graduated students as long as loan records about their books exist.

Constraints to Capture

Before drawing anything, list out the constraints hiding in the text:

- Student email uniquely identifies a student.
- Book ISBN-10 uniquely identifies a book; so does ISBN-13.
- A book may be written by **one or more** authors.
- A department belongs to **exactly one** faculty.
- Graduation date is after the joined date. (*common sense*)
- Return date of a loan is after the borrow date. (*common sense*)
- Borrow date is after the joined date.
- Copy number is consecutive, starting from 1.

Entity Sets

Nouns (not proper nouns) in the text $\Rightarrow$ entity sets:

- **Students** — “*the name and the main department of each student*”; identified by email.
- **Departments** — “*A department in NUN must belong to exactly one faculty*”
- **Faculties** — same quote as above
- **Books** — “*the title, author(s), publisher, ... for each book*”
- **Author** — “*author(s)*”
- **Copy** — “*We differentiate the copy by its copy number*”
- **Borrow** — “*the date at which a book copy is borrowed ...*”
- **Graduation / Return** — satellite entities holding the *optional* graduation date and loan-return date (see Design Notes).

Relationship Sets

Verbs / possessive phrases $\Rightarrow$ relationship sets:

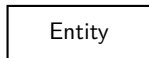
- **enroll** — Students $\leftrightarrow$ Departments: “*main department of each student*”
- **in** — Departments $\leftrightarrow$ Faculties: “*must belong to exactly one faculty*”
- **has** — Graduation $\leftrightarrow$ Students: “*If a student has graduated . . .*”
- **own** (ternary, *aggregate*) — Students (owner), Books, Copy: “*A student may own multiple copy of the same book*”
- **loan** (ternary, *aggregate*) — Students (borrower), **own**, Borrow: “*We refer to this information as a loan record*”
- **been** — **loan** $\leftrightarrow$ Return: “*the date at which it is returned*”
- **by** — Books $\leftrightarrow$ Author: “*author(s)*”

Attributes & Keys

Entity	Attributes	Key
Students	email, join, name	email
Departments	name	name
Faculties	name	name
Graduation	date	date
Books	isbn10, title (...), isbn13	isbn10 <i>and</i> isbn13
Author	name	name
Copy	copy num	copy num
Borrow	date	date
Return	date	date

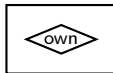
● = key attribute ○ = non-key attribute (matches the diagram notation)

ER Notation Legend



● key attribute

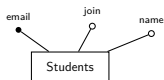
○ non-key attribute



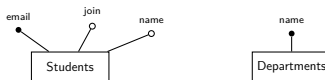
box around a relationship
= **aggregation** (it can
itself take part in another
relationship, e.g. **own** in **loan**,
loan in **been**)

Rectangle = entity set. Diamond = relationship set. ● = key attribute, ○ = non-key attribute (text sits beside the dot, connected straight to the entity/relationship). Edge labels give (min,max) participation. **own** and **loan** are *ternary* (three participants each).

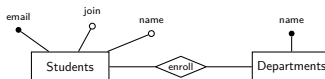
Entity-Relationship Diagram



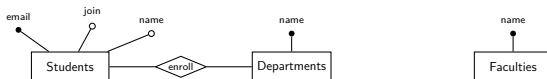
Entity-Relationship Diagram



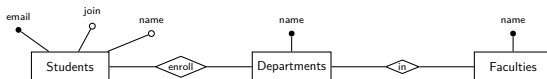
Entity-Relationship Diagram



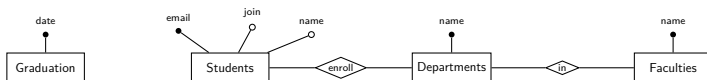
Entity-Relationship Diagram



Entity-Relationship Diagram



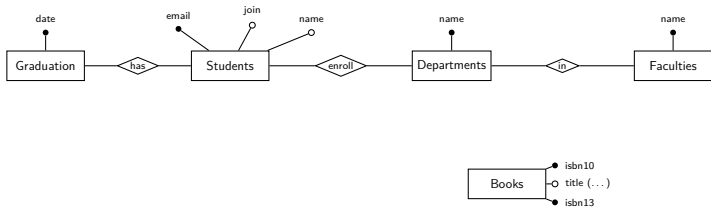
Entity-Relationship Diagram



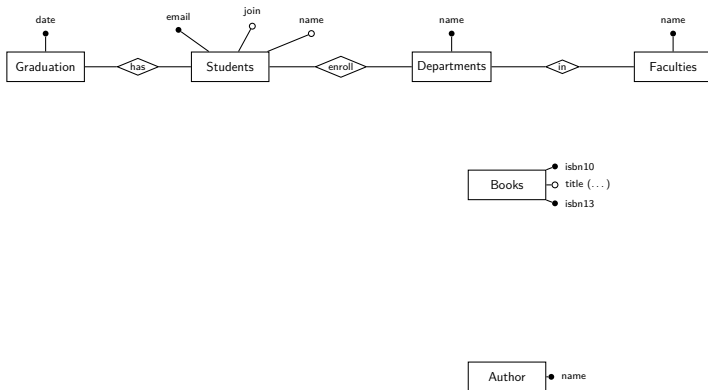
Entity-Relationship Diagram



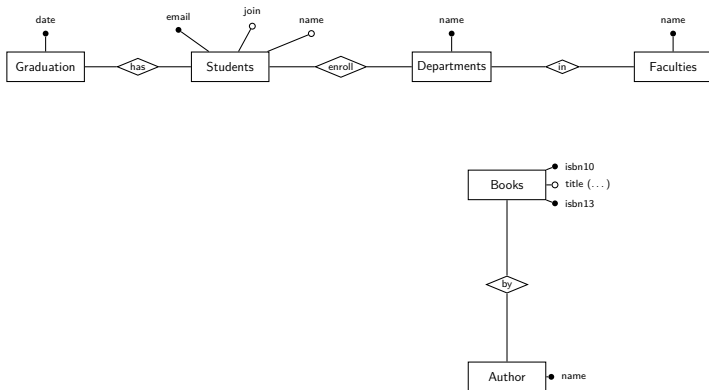
Entity-Relationship Diagram



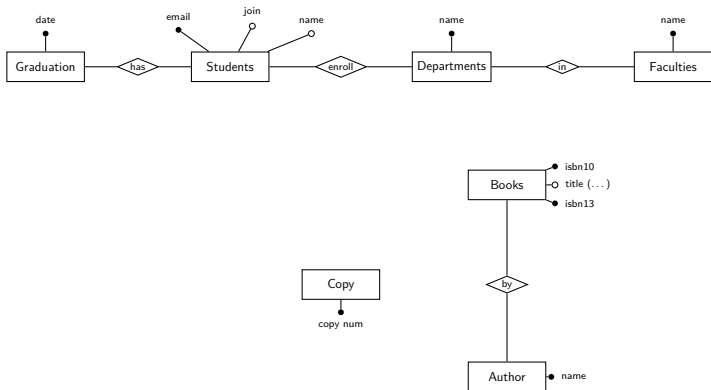
Entity-Relationship Diagram



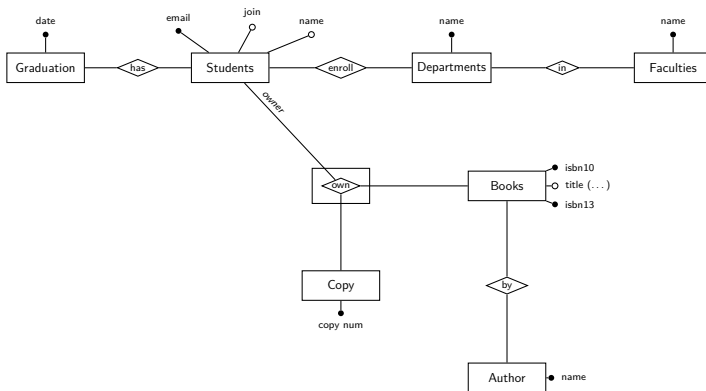
Entity-Relationship Diagram



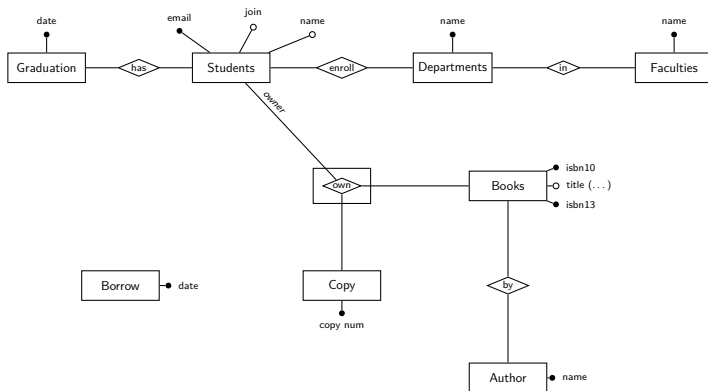
Entity-Relationship Diagram



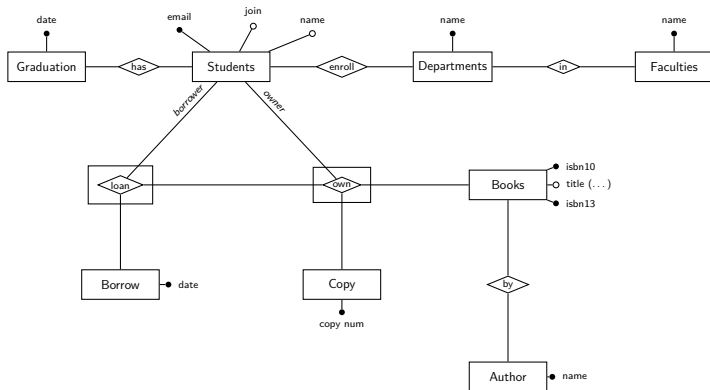
Entity-Relationship Diagram



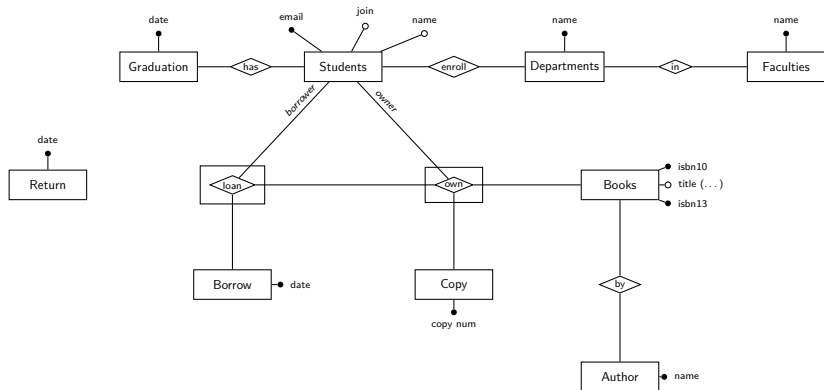
Entity-Relationship Diagram



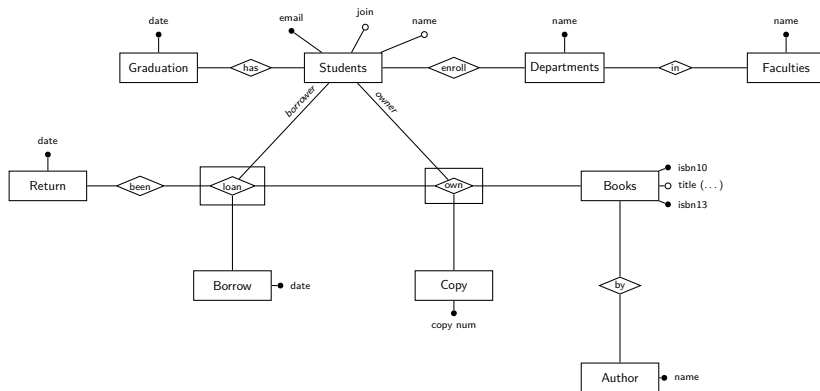
Entity-Relationship Diagram



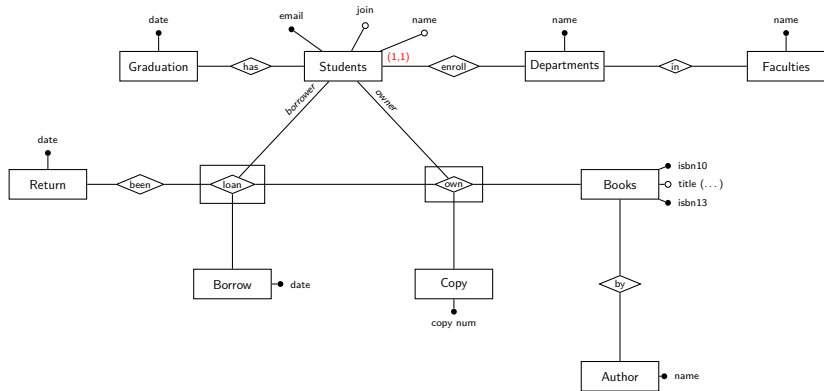
Entity-Relationship Diagram



Entity-Relationship Diagram

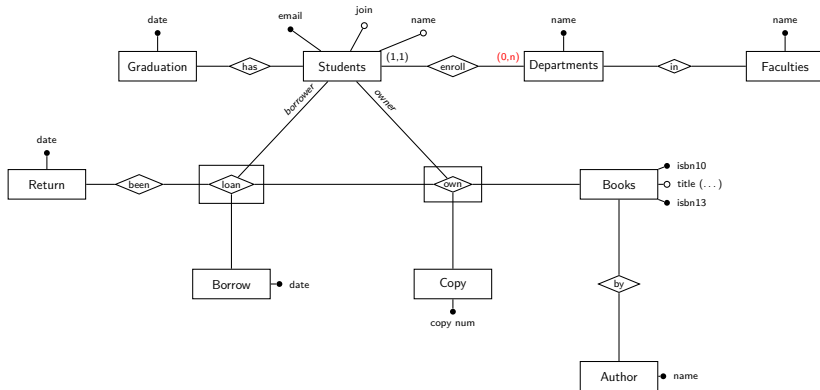


Entity-Relationship Diagram



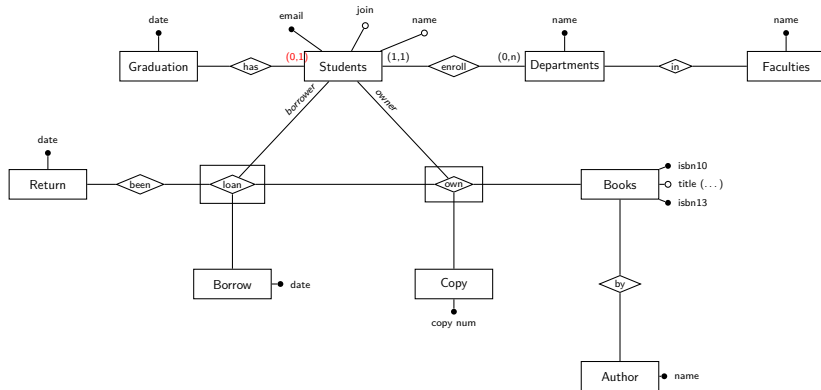
Every student enrolls in exactly one department.

Entity-Relationship Diagram



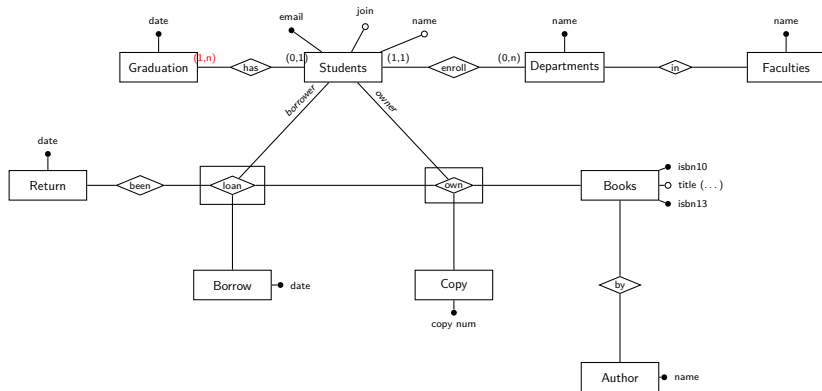
A department may have zero students.

Entity-Relationship Diagram



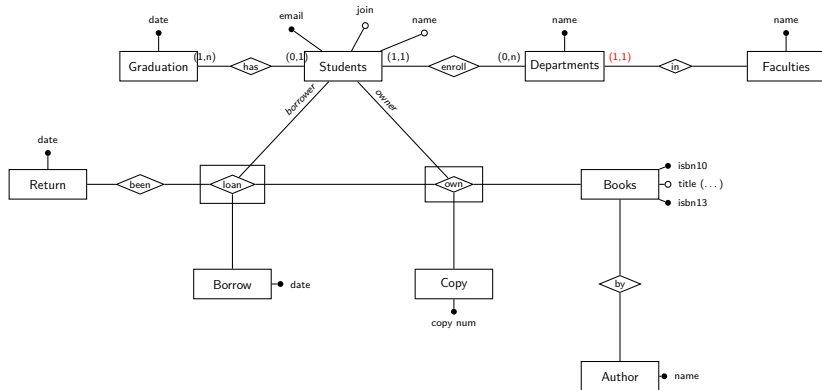
A student may or may not have graduated yet.

Entity-Relationship Diagram



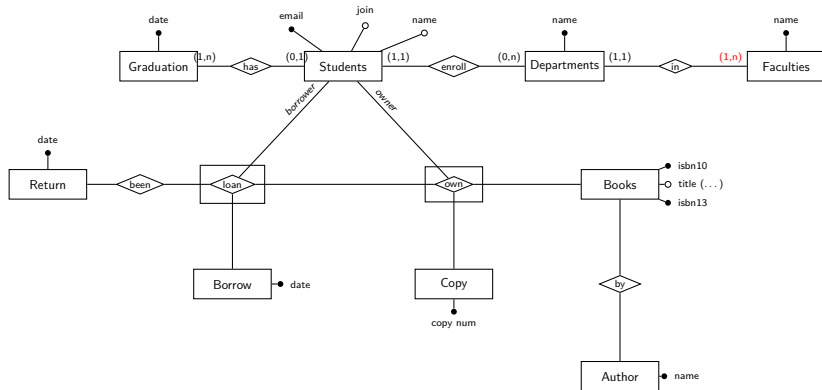
We only store a graduation date once it exists.

Entity-Relationship Diagram



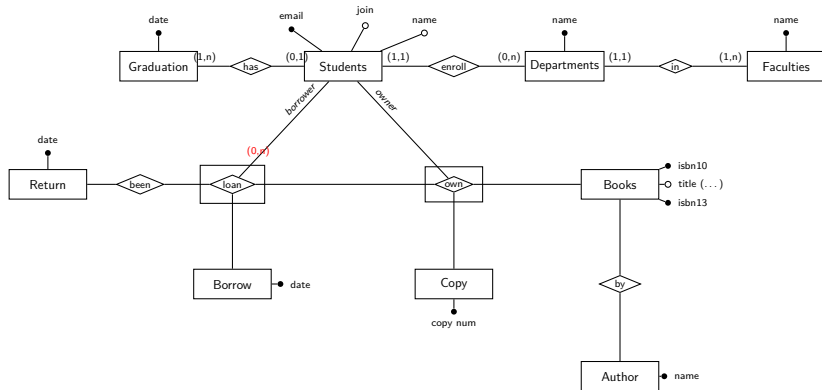
A department belongs to exactly one faculty.

Entity-Relationship Diagram



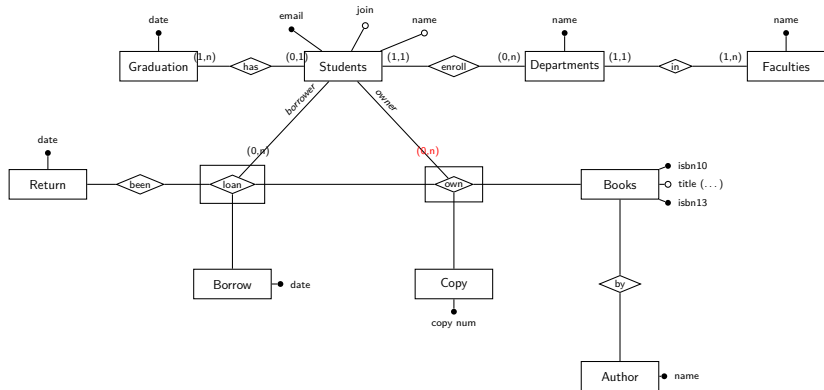
No faculty exists without at least one department.

Entity-Relationship Diagram



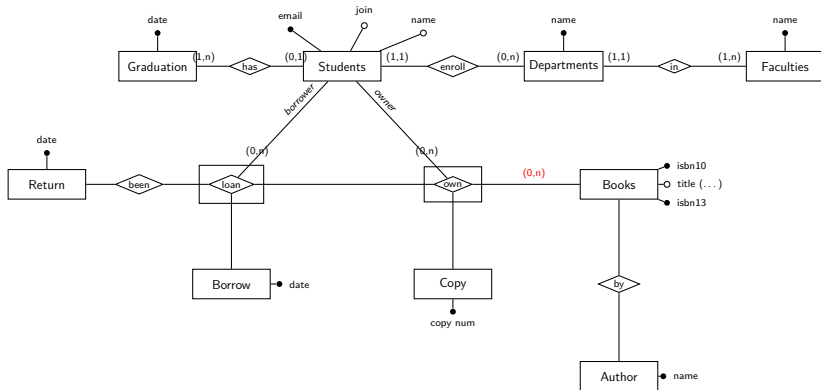
A student may hold zero or many loans.

Entity-Relationship Diagram



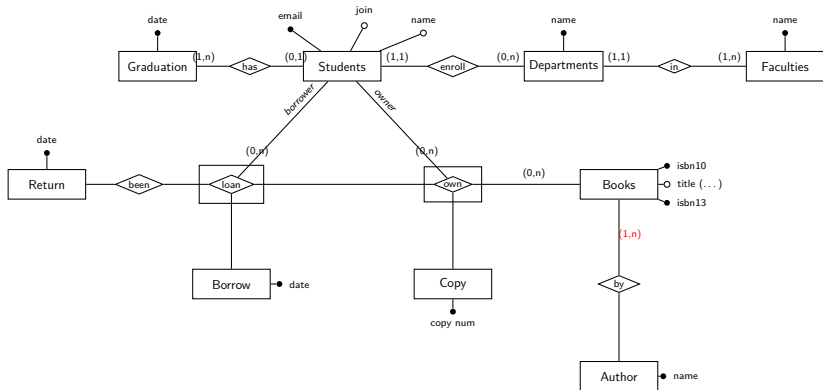
A student may own zero or many copies.

Entity-Relationship Diagram



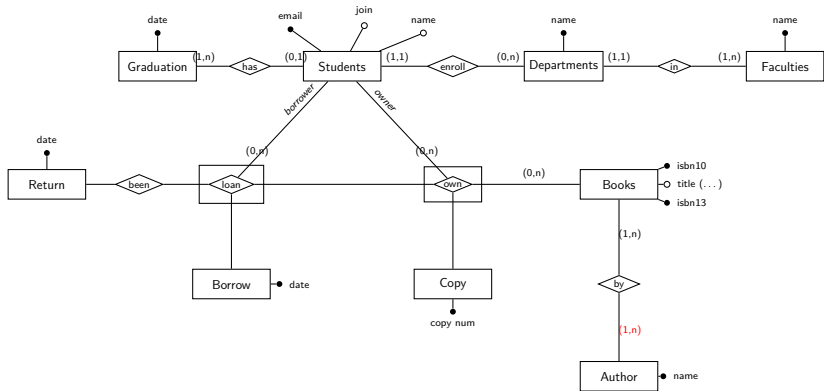
A book may exist with zero owned copies.

Entity-Relationship Diagram



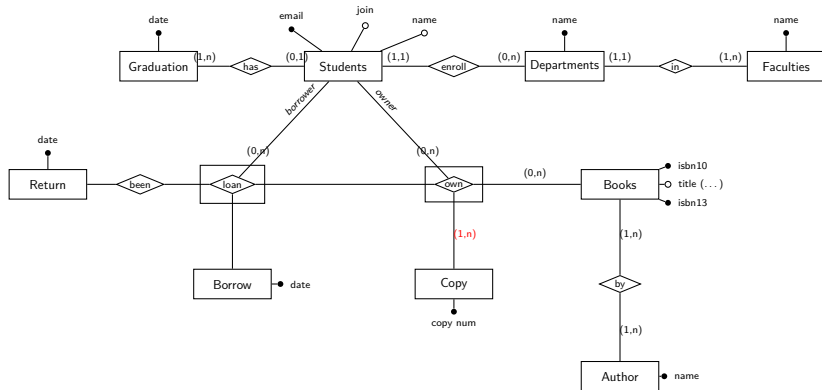
A book is written by one or more authors.

Entity-Relationship Diagram



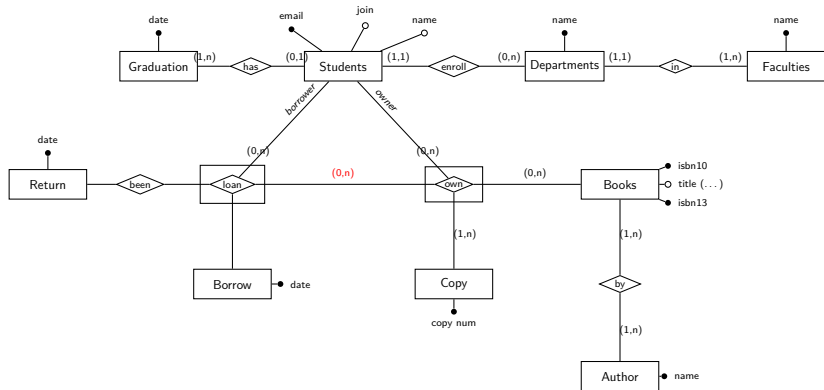
An author is only recorded once they've written a book.

Entity-Relationship Diagram



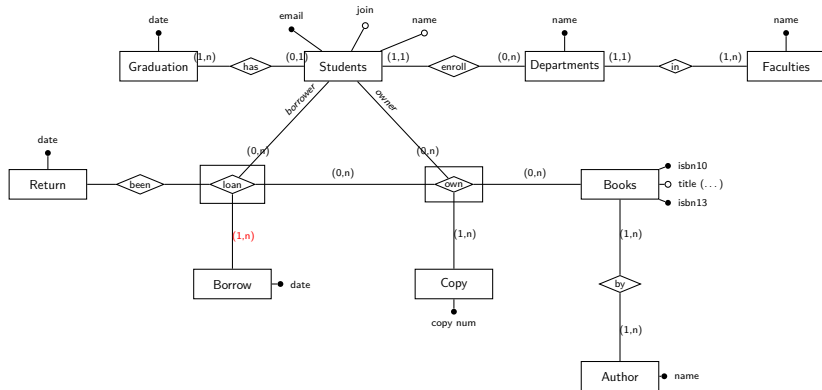
own has no meaning without a copy — forces a merge later.

Entity-Relationship Diagram



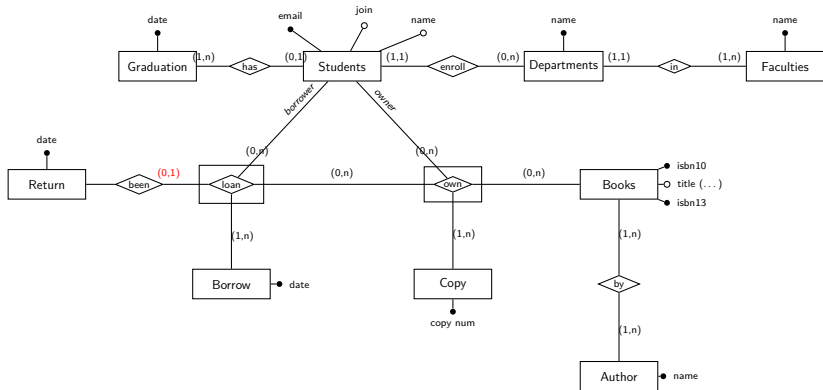
A copy may be loaned zero or many times.

Entity-Relationship Diagram



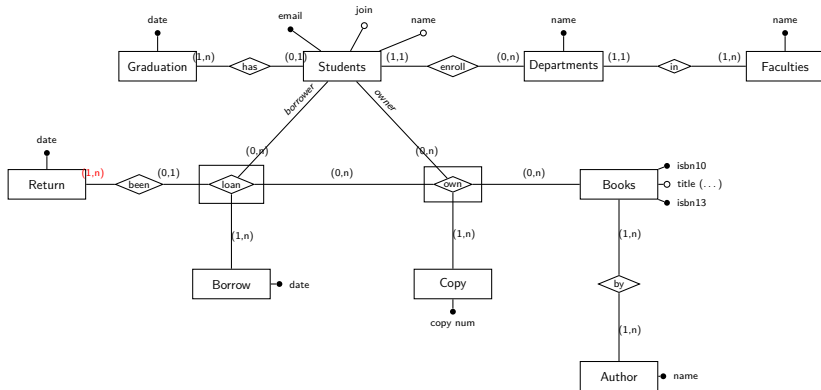
loan has no meaning without a borrow date — forces a merge later.

Entity-Relationship Diagram



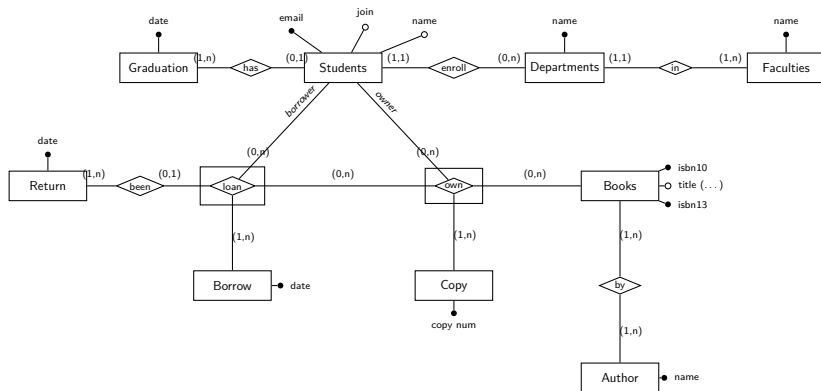
A loan may not yet have been returned.

Entity-Relationship Diagram



We only store a return date once it is actually returned.

Entity-Relationship Diagram



Design Notes — Ternary Relationships & Aggregation

- **own** and **loan** are *ternary*: a student owning/borrowing involves *three* participants at once (student, book/copy, and the copy/borrow event) — a binary relationship cannot express that.
- **Two** relationships must be boxed as **aggregates**, because each is itself a participant in a further relationship (a plain relationship cannot connect directly to another relationship):
 - **own** → boxed, because **loan** needs to reference “this particular owned copy” as a whole, not just Students or Copy alone.
 - **loan** → also boxed, because **been** needs to reference “this particular loan event” as a whole, not just Students, own, or Borrow alone.
- **Students** plays two *roles*: *owner* (into **own**) and *borrower* (into **loan**) — same entity set, two different participations.

Design Notes — Optional Attributes as Satellite Entities

- **Graduation** → **has** → **Students** is **(1,n)–(0,1)**: a student may or may not have graduated, and we don't want to store an unused graduation date on every student row. Same reasoning for **Return** → **been** → **loan**.
- These **(1,n)** satellite entities (Graduation, Return, and likewise Copy into **own**, Borrow into **loan**) are designed to be *merged* into their relationship at the logical-design stage — see next section.

Relational Schema — Department, Student

```
1 CREATE TABLE department (  
2     faculty VARCHAR(62) NOT NULL,  
3     name     VARCHAR(32) PRIMARY KEY  
4 );  
5  
6 CREATE TABLE student (  
7     name      VARCHAR(32) NOT NULL,  
8     email     VARCHAR(256) PRIMARY KEY,  
9     year      DATE NOT NULL,          -- join date  
10    graduate   DATE CHECK (graduate >= year),  
11    department VARCHAR(32) NOT NULL  
12    REFERENCES department(name)  
13 );
```

Deviation: department-faculty and student-Graduation are each merged into one table (the (1,1) side has no attributes of its own to justify a separate table; merging graduate lets us write the CHECK in one place).

Relational Schema — Book, Author

```
1 CREATE TABLE book (  
2     title      VARCHAR(256) NOT NULL,  
3     publisher  VARCHAR(64)  NOT NULL,  
4     language   VARCHAR(32)  NOT NULL,  
5     year       DATE NOT NULL,  
6     edition    INT  NOT NULL,  
7     isbn10     CHAR(10) NOT NULL UNIQUE,  
8     isbn13     CHAR(14) PRIMARY KEY  
9 );  
10  
11 CREATE TABLE author (  
12     name VARCHAR(32),  
13     book CHAR(14) REFERENCES book(isbn13),  
14     PRIMARY KEY (name, book)  
15 );
```

Relational Schema — own_copy, loan

```
1 CREATE TABLE own_copy (  
2   owner VARCHAR(256) REFERENCES student(email),  
3   book  CHAR(14) REFERENCES book(isbn13),  
4   copy  INT CHECK (copy > 0),  
5   PRIMARY KEY (owner, book, copy)  
6 );  
7  
8 CREATE TABLE loan (  
9   borrower VARCHAR(256) REFERENCES student(email),  
10  owner     VARCHAR(256),  
11  book      CHAR(14),  
12  copy      INT,  
13  borrowed  DATE,  
14  returned  DATE CHECK (returned >= borrowed),  
15  FOREIGN KEY (owner, book, copy)  
16    REFERENCES own_copy(owner, book, copy),  
17  PRIMARY KEY (borrowed, borrower, owner, book, copy)  
18 );
```

Constraints Not Captured by the DDL

- **Copy number is consecutive from 1** — needs the values across *all* rows for an (owner, book); a single-row CHECK cannot see other rows.
- **Borrow date is after the joined date** — compares columns across loan and student; a single-row CHECK cannot reach into another table.
- `graduate >= year` and `returned >= borrowed` **are** captured, via same-row CHECK constraints.
- **Audit retention** (keep books/copies/students while referenced by a copy or loan) is captured *implicitly*: the default foreign-key behaviour (ON DELETE RESTRICT) blocks any DELETE that would orphan an `own_copy` or `loan` row.

Questions?

Drop a mail at: pratik.karmakar@u.nus.edu